Topic: Chemical compounds

Question: Beryllium fluoride has the molecular formula BeF_2, and it has a molar mass of about 47 g/mol. If the molar mass of beryllium is 9 g/mol and the molar mass of fluorine is 19 g/mol, find the mass of a beryllium fluoride sample containing 228 g of fluorine.

Answer choices:

A 210 g

B 242 g

C 182 g

D 282 g

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Solution: D

Based on its molecular formula, one molecule of beryllium fluoride contains 1 beryllium atom and 2 fluorine atoms. Since fluorine has a molar mass of 19 g/mol, one mole of beryllium fluoride contains

$$2(19) = 38 \text{ g}$$

of fluorine. Therefore, since the molar mass of beryllium fluorine is 47 g, we can say that the ratio of the molar mass of the fluorine in any sample of beryllium fluoride to the total molar mass of that sample is

$$\frac{38 \text{ g}}{47 \text{ g}}$$

If we want the total mass x of a sample of beryllium fluoride that contains 228 g of fluorine, we can set up a proportion.

$$\frac{38 \text{ g}}{47 \text{ g}} = \frac{228 \text{ g}}{x}$$

$$\frac{38}{47} = \frac{228 \text{ g}}{x}$$

Cross multiplying, we get

$$38x = (228 \text{ g})(47)$$

$$38x = 10{,}716 \text{ g}$$

$$x = 282 \text{ g}$$

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Topic: Chemical compounds

Question: What is the molar mass of H_2O, if the molar mass of hydrogen is 1.00794 g/mol and the molar mass of oxygen is 15.9994 g/mol?

Answer choices:

A 18.01528

B 18.1

C 19.25672

D None of these

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Solution: A

Because one molecule of H_2O has two atoms of hydrogen and one atom of oxygen, the mass of one mole of H_2O is

$$2(1.00794) + 15.9994$$

$$2.01588 + 15.9994$$

$$18.01528 \text{ g}$$

So the molar mass of H_2O is 18.01528 g/mol.

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Topic: Chemical compounds

Question: Glucose, $C_6H_{12}O_6$, has a molar mass of 180.156 g/mol. What is the mass of oxygen in a 100 g glucose sample given that the molar mass of carbon is 12.01 g/mol and the molar mass of hydrogen is 1.00794 g/mol.

Answer choices:

A 4.0318

B 17.7622

C 32.0635

D 53.2875

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Solution: D

A molecule of glucose has 6 carbon, 12 hydrogen, and 6 oxygen atoms. We first need to find the molar mass of oxygen in a glucose molecule. Since we're only given the molar masses of carbon and hydrogen molecules, we can calculate the molar mass of the oxygen as

$$6(12.01) + 12(1.00794) + x = 180.156$$

$$x = 96.0007$$

The molar mass of oxygen in a glucose molecule is 96.0007 g/mol.

Now we can set up a proportion which consists of two ratios. The first is the ratio of the molar mass of oxygen in a glucose molecule to the molar mass of glucose, and the second is the ratio of the mass of oxygen, which we would denote as x, to the given mass of the glucose sample. Therefore, we have

$$\frac{96.0007}{180.156} = \frac{x}{100}$$

$$180.156(x) = 100(96.0007)$$

$$x \approx 53.2875 \text{ g}$$

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